

SQL Data Warehouse Project Report

1. Project Overview

This project demonstrates the design and implementation of a modern SQL Data Warehouse using PostgreSQL. The objective was to transform raw business data from multiple source systems into a structured analytical model that supports business intelligence and reporting.

The solution follows the Medallion Architecture approach consisting of Bronze, Silver, and Gold layers. Data is extracted from CRM and ERP source systems, cleansed and standardized in the Silver layer, and transformed into business-ready dimensions and fact tables in the Gold layer.

2. Business Problem

Organizations often receive data from multiple systems that contain inconsistencies, duplicate records, missing values, and varying formats. Such issues make reporting unreliable and hinder decision-making.

The objective of this project is to:

- Consolidate data from multiple source systems.
 - Improve data quality through validation and cleansing.
 - Create a scalable data warehouse architecture.
 - Provide a trusted source of data for analytics and reporting.
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3. Technology Stack

Database

- PostgreSQL

Data Processing

- SQL

Documentation

- Notion
- GitHub

Data Visualization

- Power BI

Version Control

- Git
 - GitHub
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4. Project Architecture

The project follows a three-layer architecture:

Bronze Layer

Raw data ingestion layer.

Responsibilities:

- Load source data without transformations.
- Preserve original records.
- Maintain historical source information.

Silver Layer

Data cleansing and transformation layer.

Responsibilities:

- Remove duplicates.
- Standardize formats.
- Handle null values.
- Apply business rules.
- Perform data quality validation.

Gold Layer

Business-ready analytical layer.

Responsibilities:

- Create dimension tables.
- Create fact tables.
- Support reporting and dashboarding.

Data Flow:

Source Files → Bronze Layer → Silver Layer → Gold Layer → Power BI Dashboard

5. Source Systems

CRM Source

Files:

- cust_info.csv
- prd_info.csv
- sales_details.csv

Contains:

- Customer information
- Product information
- Sales transactions

ERP Source

Files:

- cust_az12.csv
- loc_a101.csv
- px_cat_g1v2.csv

Contains:

- Customer demographics
 - Geographic information
 - Product category information
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6. Data Modeling

Dimension Tables

dim_customers

Contains:

- Customer Key
- Customer ID
- Customer Number
- Customer Name
- Country
- Marital Status
- Gender
- Birth Date

dim_products

Contains:

- Product Key
- Product ID
- Product Number
- Product Name
- Category
- Subcategory
- Product Cost
- Product Line

Fact Table

fact_sales

Contains:

- Order Number
 - Product Key
 - Customer Key
 - Order Date
 - Ship Date
 - Due Date
 - Sales Amount
 - Quantity
 - Price
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7. Data Quality Framework

Quality checks were implemented at both Silver and Gold layers.

Silver Layer Checks

Customer Validation

- Duplicate customer IDs
- Null customer IDs
- Leading/trailing spaces
- Marital status consistency

Product Validation

- Duplicate product IDs
- Null product IDs
- Negative product costs
- Invalid product dates

Sales Validation

- Invalid dates
- Incorrect sales calculations
- Invalid quantities
- Invalid prices

ERP Validation

- Invalid birth dates
- Country standardization
- Category consistency

Gold Layer Checks

Dimension Validation

- Duplicate customer keys
- Duplicate product keys

Referential Integrity

- Fact table records without matching dimensions
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8. Key SQL Techniques Used

- Common Table Expressions (CTEs)
 - Window Functions
 - ROW_NUMBER()
 - CASE Statements
 - COALESCE()
 - JOIN Operations
 - Data Quality Validation Queries
 - View Creation
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9. Project Outcomes

The project successfully:

- Integrated CRM and ERP data sources.
 - Implemented a multi-layer warehouse architecture.
 - Improved data consistency and reliability.
 - Created analytical dimension and fact models.
 - Established automated quality validation procedures.
 - Produced reporting-ready datasets for Power BI.
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10. Future Enhancements

Potential improvements include:

- Incremental data loading.
 - Automated ETL scheduling.
 - Change Data Capture (CDC).
 - Data lineage tracking.
 - Advanced monitoring dashboards.
 - Cloud deployment using AWS or Azure.
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11. Conclusion

This project demonstrates the complete lifecycle of building a SQL Data Warehouse, from raw data ingestion to business-ready analytics. By implementing Bronze, Silver, and Gold layers, the

solution provides a scalable and maintainable foundation for business intelligence and decision-making.